

PAYSON JOHANSON

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EDUCATION

Clark University, Worcester, MA

Expected: May 2026

B.A. Mathematics, Minor: Philosophy

GPA: 3.70

Awards: Dean's List First Honors, Phi Sigma Tau (Philosophy Honors Society)

RELEVANT COURSEWORK

Linear Algebra • Probability & Statistics • Real Analysis • Measure Theory • Topics in Statistics • Data Structures • Differential Equations • Genetics • Chemistry I & II • Linear Mathematical Optimization • Multivariate Calculus • Genome Project

SKILLS

Technical: Python, R, Java, Bash, Bioconda Packages

Analytical: Bioinformatics, Statistical Modeling, Computational Biology, Data Analysis, Machine Learning

WORK AND RESEARCH EXPERIENCE

Lab Member — Tabima Lab, Clark University

October 2025 – Present

- Conduct phylogenetic and comparative genomics analyses of *Basidiobolus* species using HPC cluster
- Develop automated Bash/SLURM pipelines integrating MAFFT, IQ-TREE, BUSCO, ASTER for genome analysis
- Apply Bayesian inference and ML methods to assess phylogenetic relationships and genome quality

Research Intern — Carnegie Mellon University / UnitedHealth Group

June – July 2025

- Applied statistical and ML methods in R to analyze healthcare disparities using county-level data
- Implemented Random Forest and Negative Binomial models; found education/unemployment were stronger predictors of preventable hospitalizations in POC-majority counties than provider supply

KEY PROJECTS

Comparative Phylogenomics of Basidiobolus

- Performed large-scale phylogenetic analyses on 71 genomes to investigate evolutionary relationships
- Generated ML and Bayesian trees using ITS and BUSCO gene datasets with concatenated/coalescent methods
- Automated ortholog extraction and tree-building workflows ensuring reproducibility

Healthcare Access and Racial Disparities Analysis

- Investigated racial disparities in preventable hospitalizations using CHRR county-level data
- Co-authored published report presented to faculty and healthcare analytics professionals